

GameFi Economy - Option B

Full-stack decentralized protocol for game items, resources, rentals, loot, and DAO governance.

Built with Foundry, React, The Graph, Chainlink-compatible mocks, and L2 deployment scripts.

Team split: game mechanics, economy/governance, integration/delivery.

Problem and user flows

Players need resource swaps, crafting, loot boxes, and item rentals.

Liquidity providers earn exposure through LP shares.

DAO voters govern recipes, drop rates, treasury, and upgrades.

System architecture

ERC-1155 item economy: GameItems1155.

ERC-20 resources: GOLD and CRYSTAL.

AMM, rental vault, ERC-4626 treasury, UUPS parameter proxy.

The Graph indexes events; frontend reads indexed data and contracts.

Game mechanics

Crafting burns resource tokens and mints ERC-1155 items.

LootDrop burns loot boxes and uses VRF-compatible randomness.

RentalVault escrows items and uses pull-payment earnings.

Economy layer

AMMPool is a from-scratch $x*y=k$ pool with 0.3% fee.

LP token is built into the AMM contract.

GameVault4626 acts as Timelock-controlled treasury vault.

PriceFeedAdapter rejects stale or invalid Chainlink prices.

Governance

GameToken is ERC20Votes + ERC20Permit.

Governor settings: 1-day delay, 1-week period, 4% quorum, 1% proposal threshold.

Timelock delay: 2 days.

Lifecycle test demonstrates propose -> vote -> queue -> execute.

Advanced Solidity

UUPS upgrade: GameParametersV1 to GameParametersV2.

Factory uses CREATE and CREATE2.

AMMMath includes Solidity and inline Yul quote paths for benchmarking.

Security design

AccessControl on privileged functions.

ReentrancyGuard and CEI on external transfer paths.

SafeERC20 for ERC-20 movements.

No tx.origin, no transfer/send, no block values as randomness.

Testing strategy

88 test functions plus 6 invariants.

Unit, fuzz, invariant, fork, and vulnerability case-study tests.

Fuzz coverage includes AMM swaps, vault operations, and voting power.

Invariants include k, reserves, LP supply, treasury accounting, and vault rounding.

Frontend and subgraph

Frontend supports wallet connection, network detection, readable errors, balances, votes, reserves, vault shares.

State-changing actions: swap, craft, deposit, delegate, vote.

Subgraph indexes players, crafts, items, swaps, rentals, proposals, and votes.

Deployment and demo

Deploy with script/Deploy.s.sol to Base Sepolia, Arbitrum Sepolia, or Optimism Sepolia.

Run VerifyPostDeploy to check Timelock and Governor settings.

Fill deployment JSON, frontend env, subgraph addresses, gas table, and final explorer links.